

I. Introduction

- Perspectives on the development of AI
- Fundamentals - Iterative Guessing, **Cost Functions, Gradient Descent**, Regression vs Classification, Logistic Regression, Binary and Multi-classification

II. Neural Network Basics

- Fundamentals - Perceptron and Deep Neural Networks, Neural Networks vs Linear Models, NN as Function Approximator, Backpropagation, Framework for Training
- **Training - Gradient Descent/Ascent**, SGD and Mini-batch SGD, Vanishing Gradient Problem, Not Zero-Centred Problem, Choice of Activation Function, Learning Rate Problems, Momentum and Nesterov Momentum, Adaptive Learning Rates, **Overfitting Problem, Weight Regularization, Dropout, Early Stopping**, Network Distillation, Batch Normalization

III. Vision

- **Vision and CNN**
- ImageNet: AlexNet 2012, ZFNet 2013, VGGNet 2014, GoogLeNet 2014, ResNet 2015, ReduNet 2022

- **Visualizing and Understanding CNNs, Adversarial Images**
- Applications: Object Localization, Object Detection, Semantic Segmentation

IV. Deep Generative Models

- **DeepDream**, Style Transfer
- Autoencoder, Variational Autoencoder
- **GAN**, Conditional-GAN, Pix2Pix, CycleGAN

V. Natural Language Processing

- RNN and NLP tasks, LSTM
- **Word embedding : CBOW, Skip-gram, Elmo**
- MT, Encoder-Decoder, Seq2Seq
- **Attention, Transformer and BERT**
- **GPT and LLMs**
- More on LLMs: Size, Pruning and distillation, Benchmark metrics, LoRA fine-tuning and quantization, **Prompting**
- RL and DeepSeek

VI. Ethics

Wednesday, 10 Decemebr 2025 (6:30-8:30PM)

SEVEN Questions – all compulsory

CLOSED BOOK / NO NOTES

BRING CALCULATOR

SHORT ANSWERS

Student Feedback on Teaching and Learning (SFTL)

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- There are separate forms for the course, professor and tutor.
- Your evaluation will be saved anonymously, without any identification.
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**Remember to give your feedback
for improving our course**